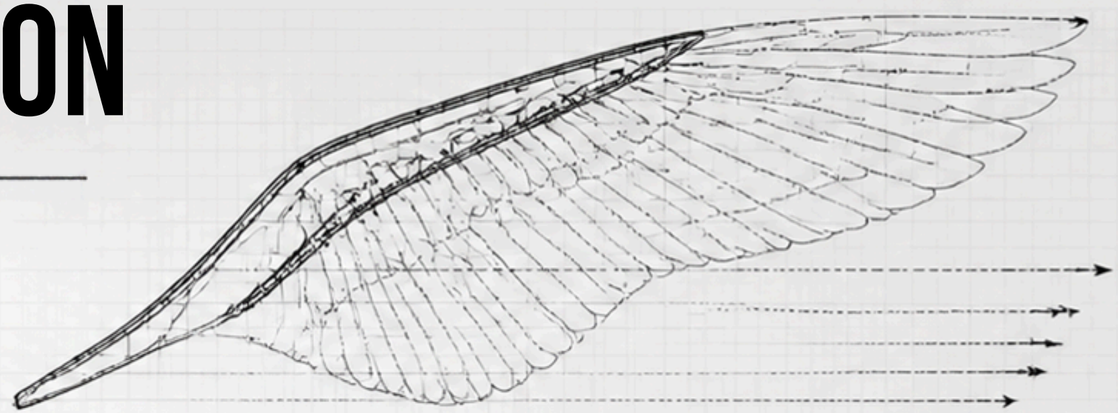
RESEARCH & DOCUMENTATION

BIRD-INSPIRED MORPHING WINGS



This project translates observations of avian flight into engineering principles for adaptive and compliant morphing wings. Research was organized across three stages: biological observations, engineering applications, and experimental evidence. These findings informed the development of our own digital morphing-wing model.



01 BIOLOGICAL OBSERVATIONS

- **Owl feathers:** Serrated leading edges, soft surfaces, and fringed trailing edges contribute to quieter airflow.
- **Albatros wings:** Long, narrow wings and dynamic soaring support efficient flight with reduced induced drag.
- **Swift wings:** Flexible wing joints and variable sweep support agile flight across changing speeds.
- **Peregrine falcon:** Wings are drawn closer to the body during high-speed dives, reducing drag.
- **Pigeon wing mechanics:** Coordinated wrist and finger joints control feather positioning, offering inspiration for compact morphing mechanisms.

02 ENGINEERING TRANSLATION

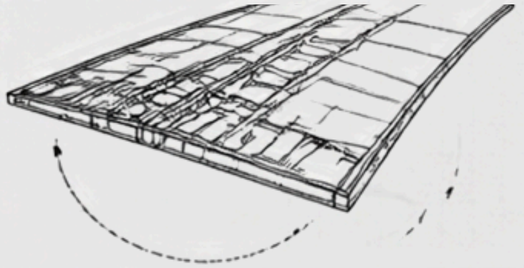
• **Continuous deformation** replaces abrupt, hinged movements to produce a smoother aerodynamic contour.
• **Camber, span, twist, and sweep** represent distinct morphing modes, each offering different aerodynamic benefits and trade-offs.
• **Compliant skins and internal ribs** must remain flexible enough to deform while maintaining sufficient structural stiffness under aerodynamic loads.
• **Closed-loop control** uses sensors and feedback to monitor wing deformation and adjust actuators accordingly.

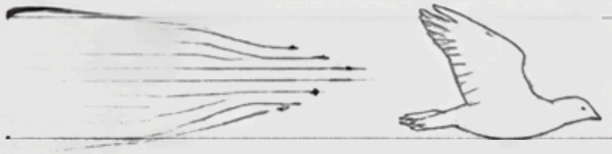
03 EVIDENCE USED IN THE PROJECT

- **Owl aerodynamics**: Clark et al. (2017) – owl-wing features for silent flight.
- **FishBAC morphing**: Woods et al. (2014) and Rivero et al. (2020) – wind-tunnel studies on compliant camber morphing.
- **Morphing-wing structures**: A Review of Modelling and Analysis of Morphing Wings (2018) – structural modelling and design challenges.
- **NASA ACTE / FlexSys**: Flight testing of an adaptive compliant trailing edge.
- **NASA X-53 and DARPA MFX**: Research into active aeroelastic control and morphing structures.
- **Airbus UpNext**: **AlbatrossONE** and Extra Performance Wing research.
- **Technical documentation**: NASA Technical Reports Server (**NTRS**) and NASA Armstrong Flight Research Center.

04 OUR DESIGN OUTPUT

• Research findings were translated into a bird-inspired morphing-wing model incorporating adaptive wing geometry and a feedback-control concept.
• Project model: An interactive digital simulation demonstrating morphing-wing behaviour and aerodynamic effects.
Live model: biomimicry-morphingwings-omega.vercel.app





ADDITIONAL VISUAL & PRESENTATION SOURCES: Images located via Google Images • Aviation Week (aircraft image, Slide 37) • Smithsonian Institution (historical aviation imagery) • Wright Brothers (Public Domain) • Bernard Spragg (CC0) • Canva (presentation design)

TRACEABILITY MAP • BIOLOGY • ENGINEERING • MODEL